

Code :

```
#include <Arduino.h>

void dangerousFunction(int size)
{
    char bigArray[size];

    for(int i = 0; i < size; i++)
    {
        bigArray[i] = i;
    }

    Serial.printf("Array Size: %d bytes -> OK\n", size);
}

void setup()
{
    Serial.begin(115200);
    delay(1000);

    Serial.println("=== Stack Overflow Test ===");

    dangerousFunction(1000);
    dangerousFunction(2000);
    dangerousFunction(5000);
    dangerousFunction(10000);
    dangerousFunction(20000);
    dangerousFunction(30000);
    dangerousFunction(40000);
    dangerousFunction(50000);
}
```

```
void loop()
```

```
{
```

```
}
```

Array Size(Bytes)	Result
1000	OK
2000	OK
5000	OK
10000	OK
20000	OK
30000	May Fail
40000	May Fail
50000	Likely Crash

Actual values depend on ESP32 configuration.